Bluestock MF Capstone Report

A final project report covering data sources, ETL design, EDA, performance analysis, and dashboard insights.

Executive Summary

This report summarizes the Bluestock MF capstone project. It explains the investment problem, data sources, ETL design, exploratory data analysis, performance analytics, dashboard insights, limitations, and recommendations. The report is intended for stakeholders seeking a data-driven mutual fund selection framework for Indian equity schemes.

Data Sources

Data sources include synthetic master datasets and live NAV feeds. The project combines AMFI scheme metadata, NAV history, portfolio holdings, SIP records, returns, AUM, benchmark data, expense ratios, fund manager details, and risk metrics. Live NAV values are fetched from MFAPI to supplement historical model outputs.

ETL Design

The ETL design follows a structured pipeline: generate base datasets, clean raw inputs, load structured tables, and produce analytics-ready outputs. Raw CSVs are transformed into normalized tables and verified for schema consistency, data quality, and AMFI scheme coverage. The pipeline supports repeatable execution and report regeneration with a single orchestration script.

Exploratory Data Analysis Findings

Exploratory Data Analysis highlights NAV distribution, volatility trends, correlation structure, and expense ratio patterns. Key findings show how liquidity, expense ratios, and sector allocation drive performance differences across large-cap schemes. Return profiles reveal that low volatility schemes tend to underperform on multi-year CAGR but provide improved risk-adjusted outcomes.

Performance Analysis

Performance analysis examines CAGR, Sharpe ratio, Sortino ratio, alpha/beta, drawdowns, and fund scorecard metrics. The analysis identifies top-performing schemes across 1-year, 3-year, and 5-year horizons and quantifies downside risk through VaR/CVaR. Sector concentration and SIP continuity metrics also inform suitability for conservative and growth-focused investors.

Dashboard Screenshots

The dashboard visualizes key insights, including follow-on charts for NAV trends, fund comparison, portfolio composition, and investor analytics. Screenshots capture the reporting experience and surface actionable findings for portfolio managers and retail investors.

Limitations

The analysis uses synthetic NAV history for model testing and a limited set of 20 sample schemes. Live NAV values are sampled from MFAPI and may not represent all plan variants or direct/regular distinctions. Future work should expand the universe, incorporate actual AMFI daily feeds, and validate with client-level SIP behavior.

Recommendations

Recommendations include focusing on funds with strong risk-adjusted returns, controlling expense ratio leakage, and monitoring sector concentration. The dashboard can be used to track fund rankings, SIP continuity, and portfolio risk metrics over time. The capstone project should evolve into an automated reporting service with scheduled data updates and an investor-facing dashboard.

Final deliverables generated by build_report.py.